

MAT1002 Final Examination

Tuesday, May 13th, 2025

Time: 13:30 PM — 16:30 PM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The maximum score of this examination is **132**.*
3. *There are **13** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Answer all questions on the **answer book**.*
6. *Show your intermediate steps **except Question 1** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *Express irrational numbers in exact forms instead of decimal forms; e.g., write $\sqrt{2}$ instead of $1.414\dots$, and write $\ln 2$ instead of $0.693\dots$*

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MAT1002 Final Exam Questions

1. [27] Short questions: no intermediate step is required.

- (i) True (T) or False (F): The 3D space without the origin is a simply connected region.
- (ii) True (T) or False (F): Suppose the function $z = f(x, y)$ is continuous with respect to x for any fixed y , and continuous with respect to y for any fixed x , then it is a continuous function.
- (iii) True (T) or False (F): The vector field $\mathbf{F} = (\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2})$ passes the component test, thus it is conservative in its domain.
- (iv) True (T) or False (F): Suppose $z = f(x, y)$ is continuous function on the region R where $R = R_1 \cup R_2$. Then we have

$$\iint_R f(x, y) \, dA = \iint_{R_1} f(x, y) \, dA + \iint_{R_2} f(x, y) \, dA.$$

- (v) Rewrite the following integral in *Cylindrical coordinates*, in the order of $dzd\theta dr$:

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{x^2+y^2} f(x, y, z) \, dz \, dy \, dx.$$

- (vi) Find the direction derivative for the function $w = xy + \cos(z) - \frac{y}{x}$ at the point $(2, 1, 0)$ in the direction $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$.
- (vii) Find the direction (the unit vector) that parallel to both the plane $3x - 2y + z = 0$ and $x + y - 3z = 0$.
- (viii) Find the quadratic approximation at $(0, 0)$ of $z = \ln(1+x) \sin(y)$.
- (ix) Compute $\sum_{n=1}^{\infty} \frac{2n+1}{n!} = (\quad)$.

2. [15] Determine the following limits. If the limit exists (as a real number), find the value and show your steps; otherwise, explain why it does not exist.

- (i) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2+y^2}$

(ii) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$

(iii) $\lim_{(r,\theta) \rightarrow (0,\pi/2)} \frac{r \cos \theta}{r^2 + 1}$

3. [6] If $f(x, y) = x \sin y + y \ln x$, find the second-order derivatives: $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y \partial x}$, $\frac{\partial^2 f}{\partial y^2}$, $\frac{\partial^2 f}{\partial x \partial y}$.

4. [6] Let Surface S be characterized by the equation $z = x^2 - y^2$. Find the tangent plane M and normal line L to the surface at point $(2, 1, 3)$.

5. [6] Find the point on the intersection line between plane $2x + y - z = 4$ and plane $x + y + z = 3$ that has minimum distance from the origin $(0, 0, 0)$

6. [8] Find and classify the critical points (as local maxima, local minima, or saddle points) of the function: $f(x, y) = x^3 + y^3 - 3xy$.

7. [6+6] Evaluate the following integrals.

(a) $\iint_R (3 - x - y) dA$, where R is the region enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

(b) $\int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} dy dx$

8. [4+4] Consider a solid bounded from below by a sphere of radius one centered at the origin and from above by the paraboloid $z = 1 - x^2 - y^2$.

(a) Express the volume of the solid in terms of a triple integral in cylindrical coordinates.

(b) Find the volume of the solid by evaluating the integral in (a).

9. [3+5] Consider the integral

$$\int_0^{2/3} \int_y^{2(1-y)} (x + 2y)e^{y-x} dx dy.$$

(a) Sketch the integration region R in the xy -plane.

- (b) Using the transformation $u = x + 2y$, $v = x - y$, express it as an integral over a suitable region G in the uv -plane, and sketch the region G (**DO NOT** evaluate the integral.)

10. [4+4+4] (a) For what values of b and c will

$$\mathbf{F} = (y^2 + 2cxz)\mathbf{i} + y(bx + cz)\mathbf{j} + (y^2 + cx^2)\mathbf{k}$$

be a gradient field?

- (b) Find a potential function of

$$\mathbf{F} = (y^2 + 4xz)\mathbf{i} + (2xy + 2yz)\mathbf{j} + (y^2 + 2x^2)\mathbf{k}$$

- (c) If $\mathbf{F}$ in (b) is a force field, find the work done by $\mathbf{F}$ on the particle that moves from the point $(0, 0, 0)$ to $(1, 2, 2)$ along the line segment joining them.

11. [8] (a) Find the flux of the fields $\mathbf{F}(x, y) = \langle x, y \rangle$ across the closed path consisting of:

1. The right semicircle from $(0, 2)$ to $(0, -2)$
2. The line segment on the y -axis from $(0, -2)$ to $(0, 2)$

- (b) Find the circulation of the fields $\mathbf{F}(x, y) = \langle -y, x \rangle$ around the closed path consisting of:

1. The right semicircle from $(0, 2)$ to $(0, -2)$. (clockwise)
2. The line segment on the y -axis from $(0, -2)$ to $(0, 2)$

12. [8] Consider the vector field $\mathbf{F}(x, y, z) = \langle x^2y, y^2z, z^2x \rangle$. Compute the outward flux of $\mathbf{F}$ through the surface of the cube bounded by the planes $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, and $z = 1$.

13. [8] Consider the vector field $\mathbf{F}(x, y, z) = \langle y^2, z^2, x^2 \rangle$. Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy -plane, oriented upward. The boundary of S is the curve C (a circle in the xy -plane). Compute the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} d\sigma$$

using Stokes' Theorem.